LAPORAN PRAKTIKUM

POSTTEST (3)

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

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Kelas (A2 ‘25)

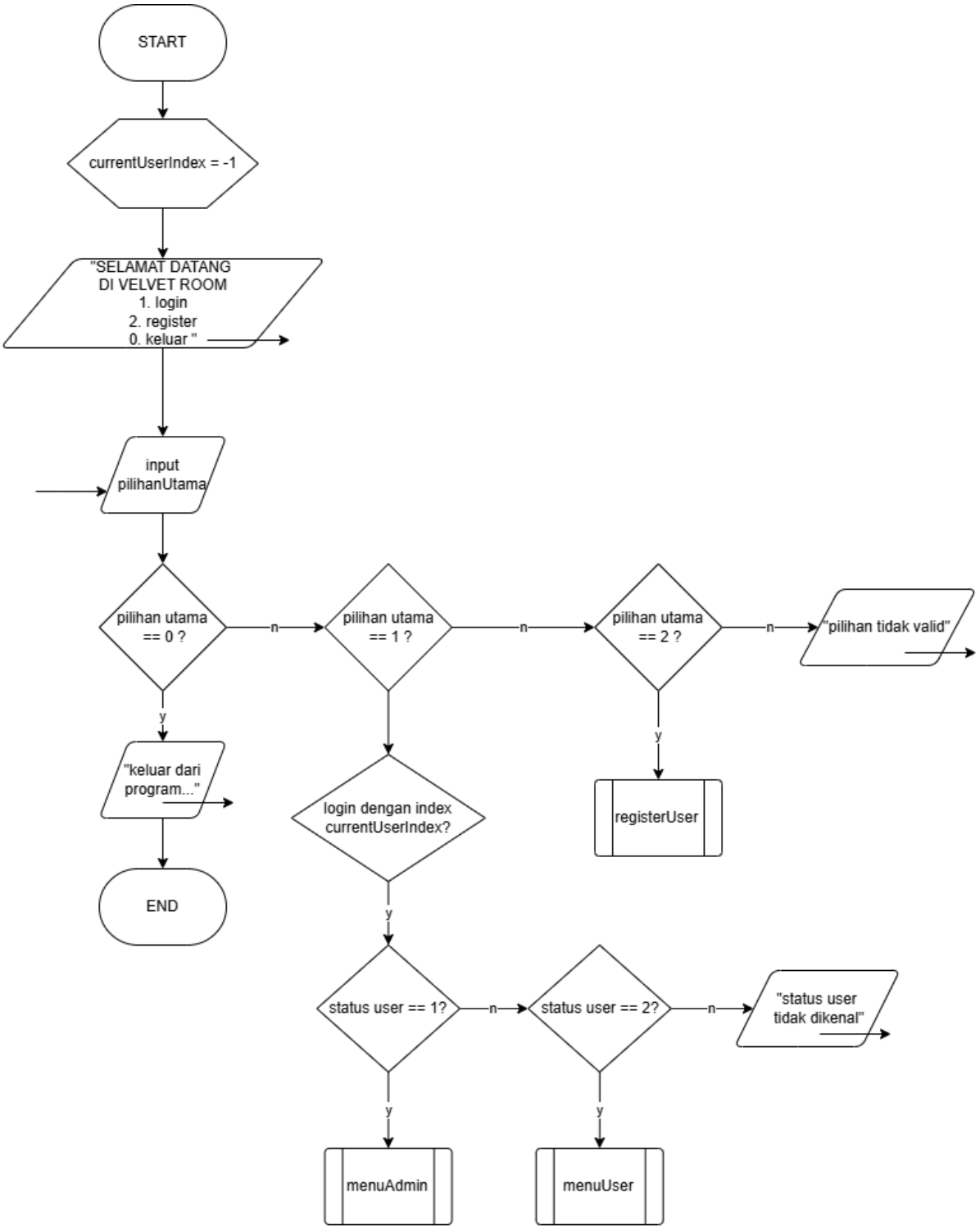
PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

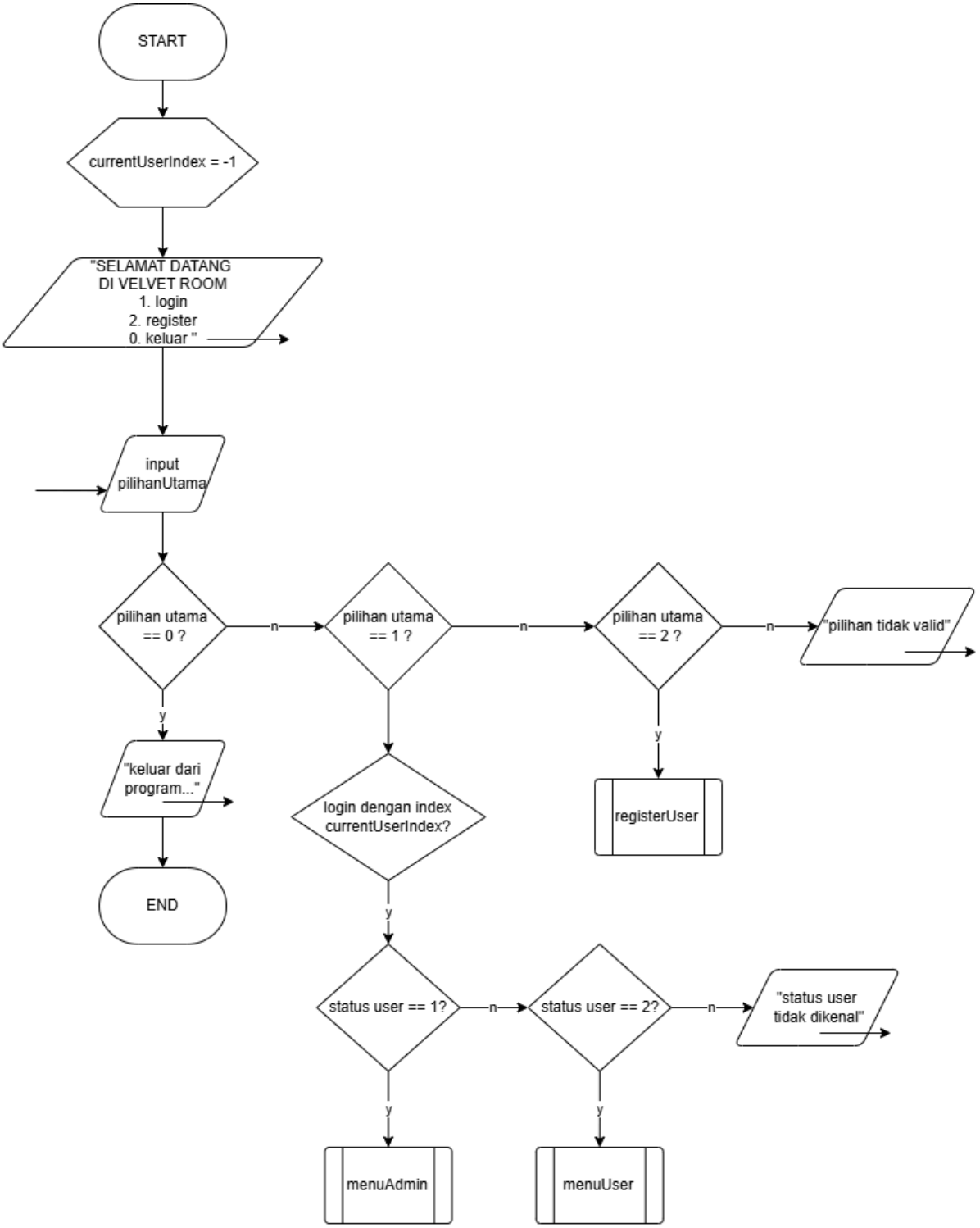
SAMARINDA

2025

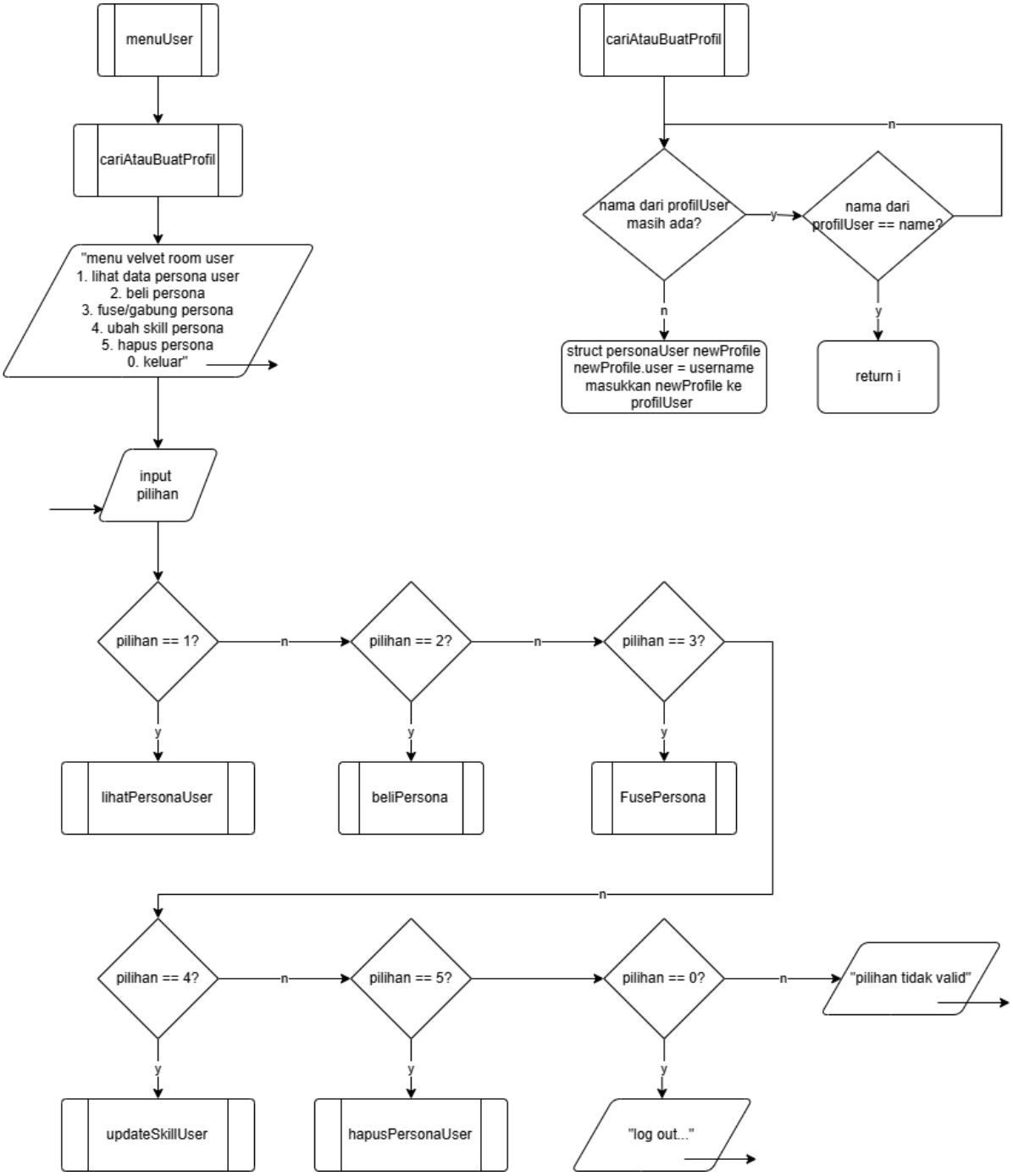
1. Flowchart



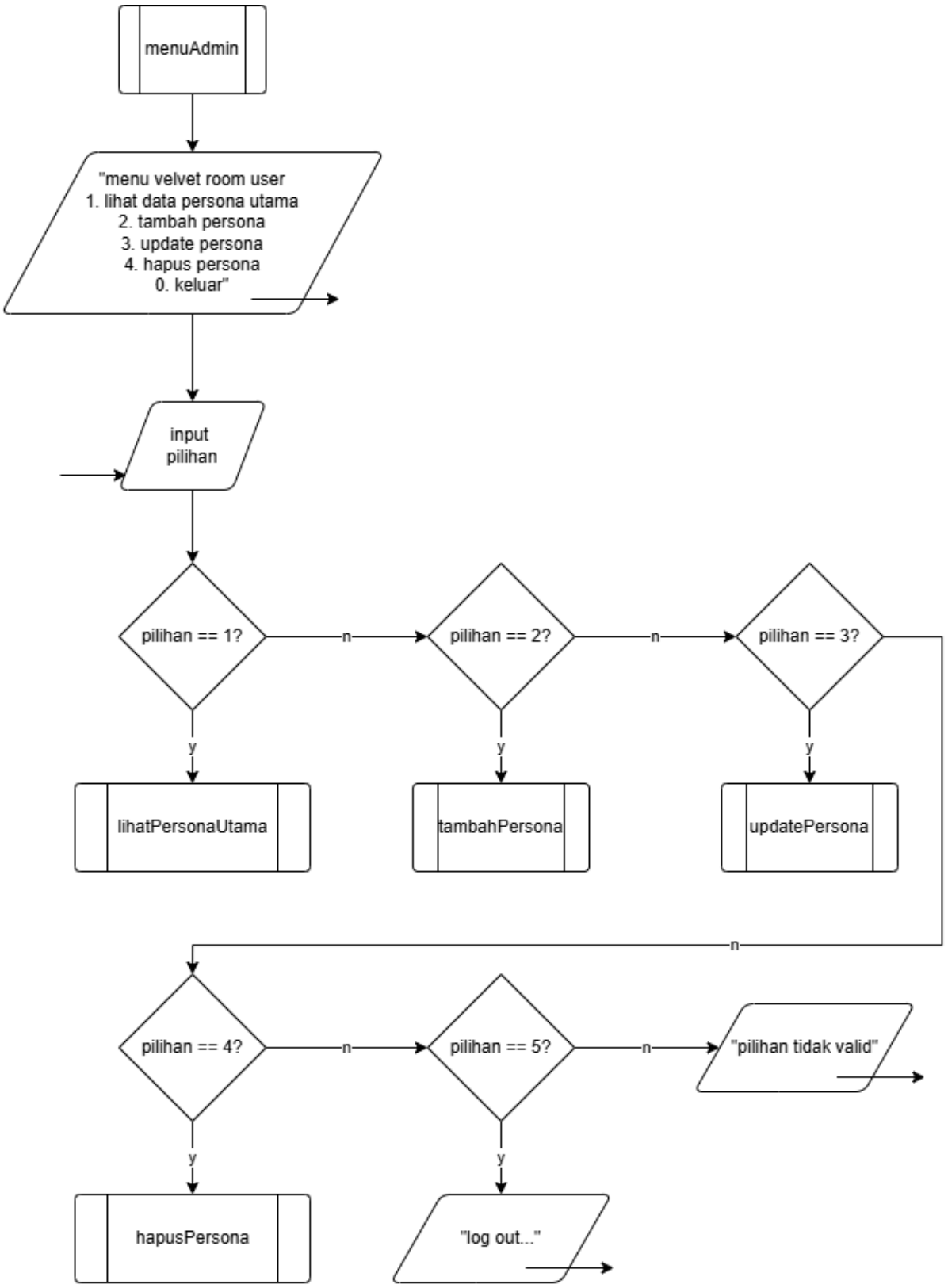
gambar 1.1 main program



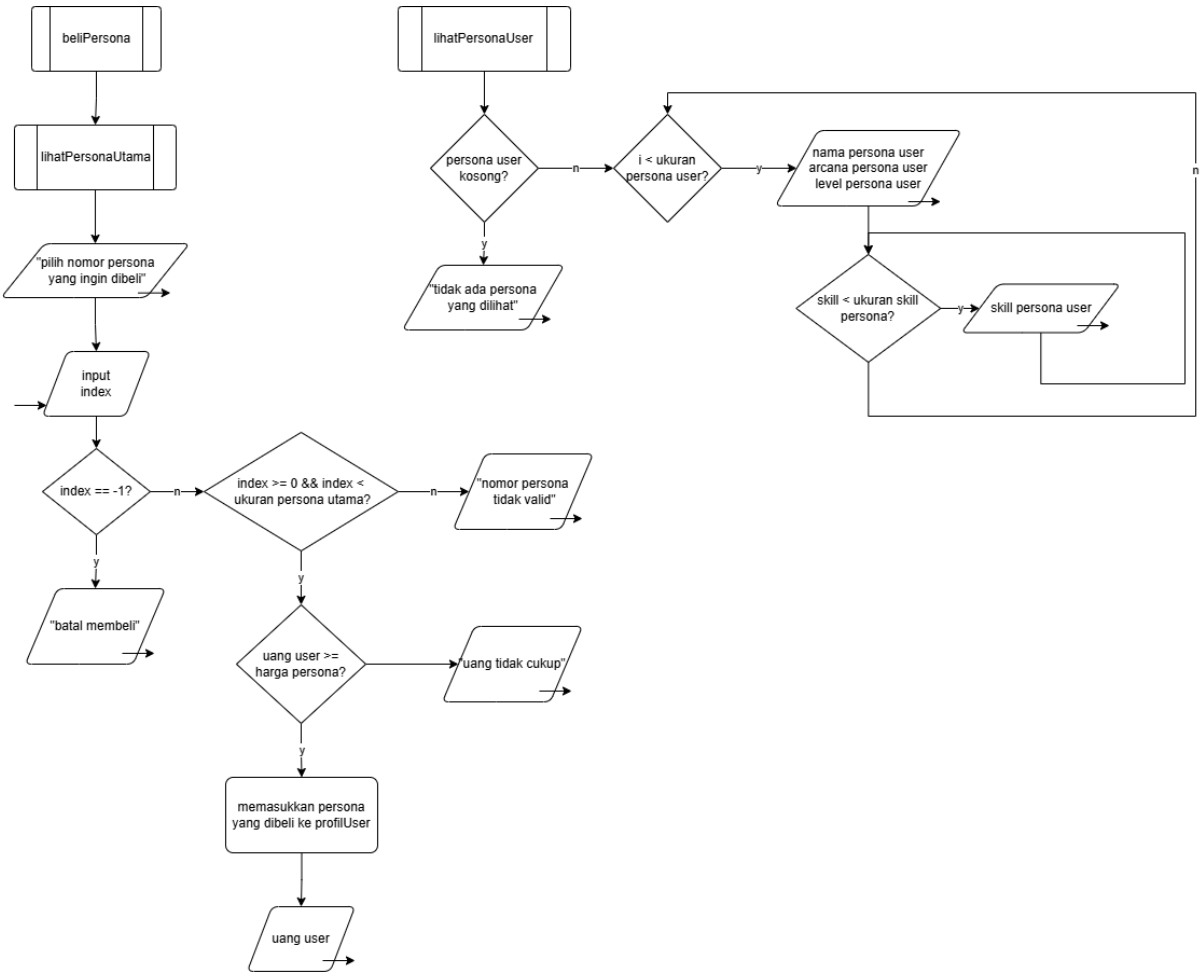
gambar 1.2 flowchart fungsi login dan fungsi register



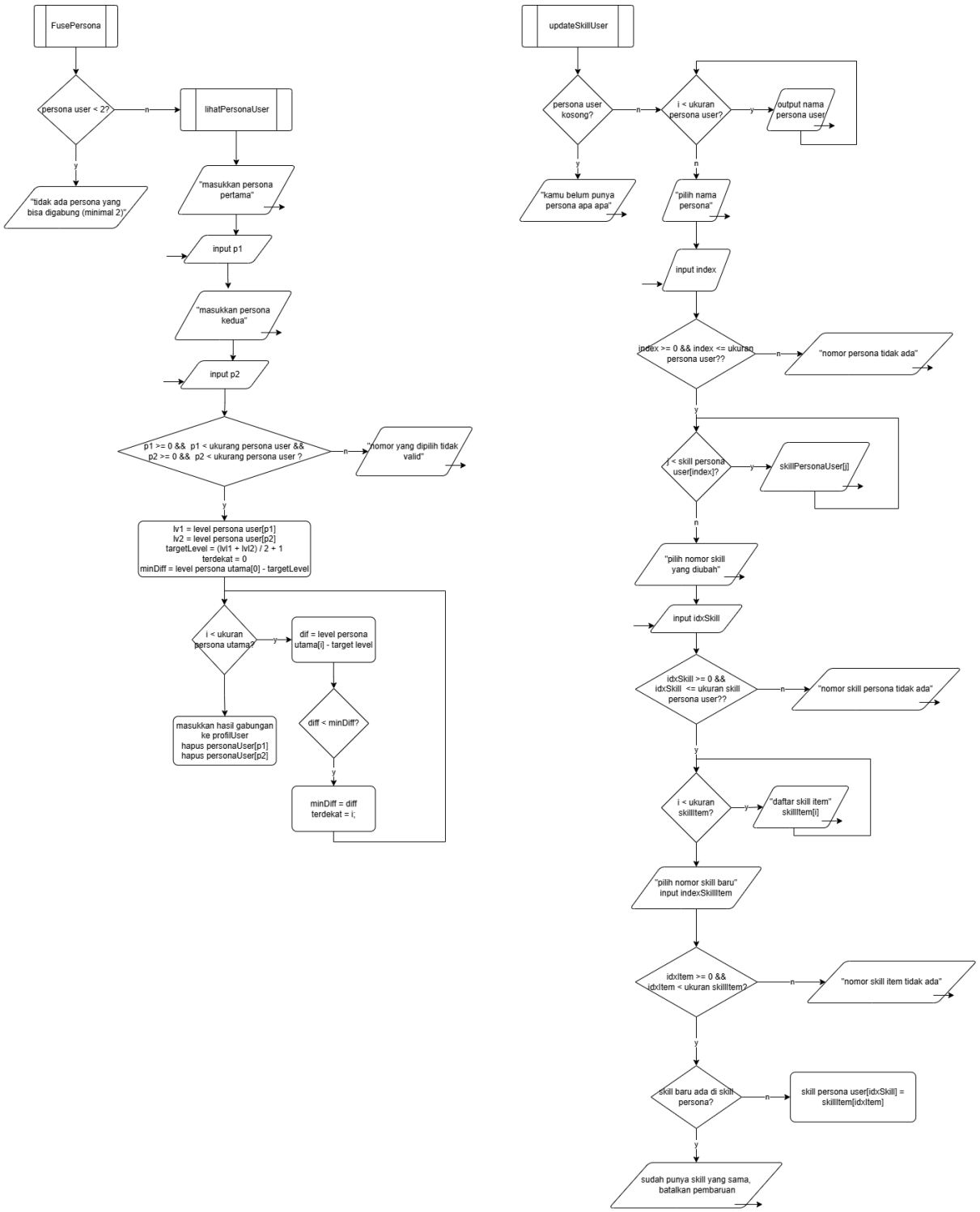
gambar 1.3 flowchart fungsi menuUser



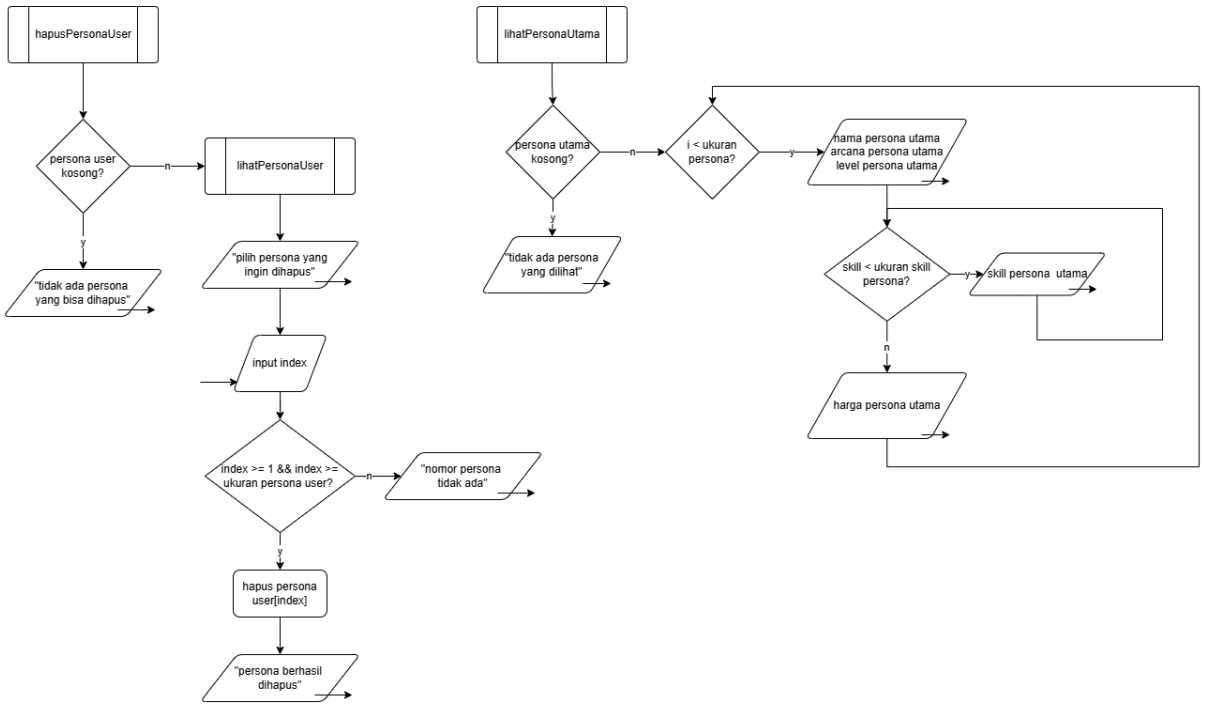
gambar 1.4 flowchart fungsi menuAdmin



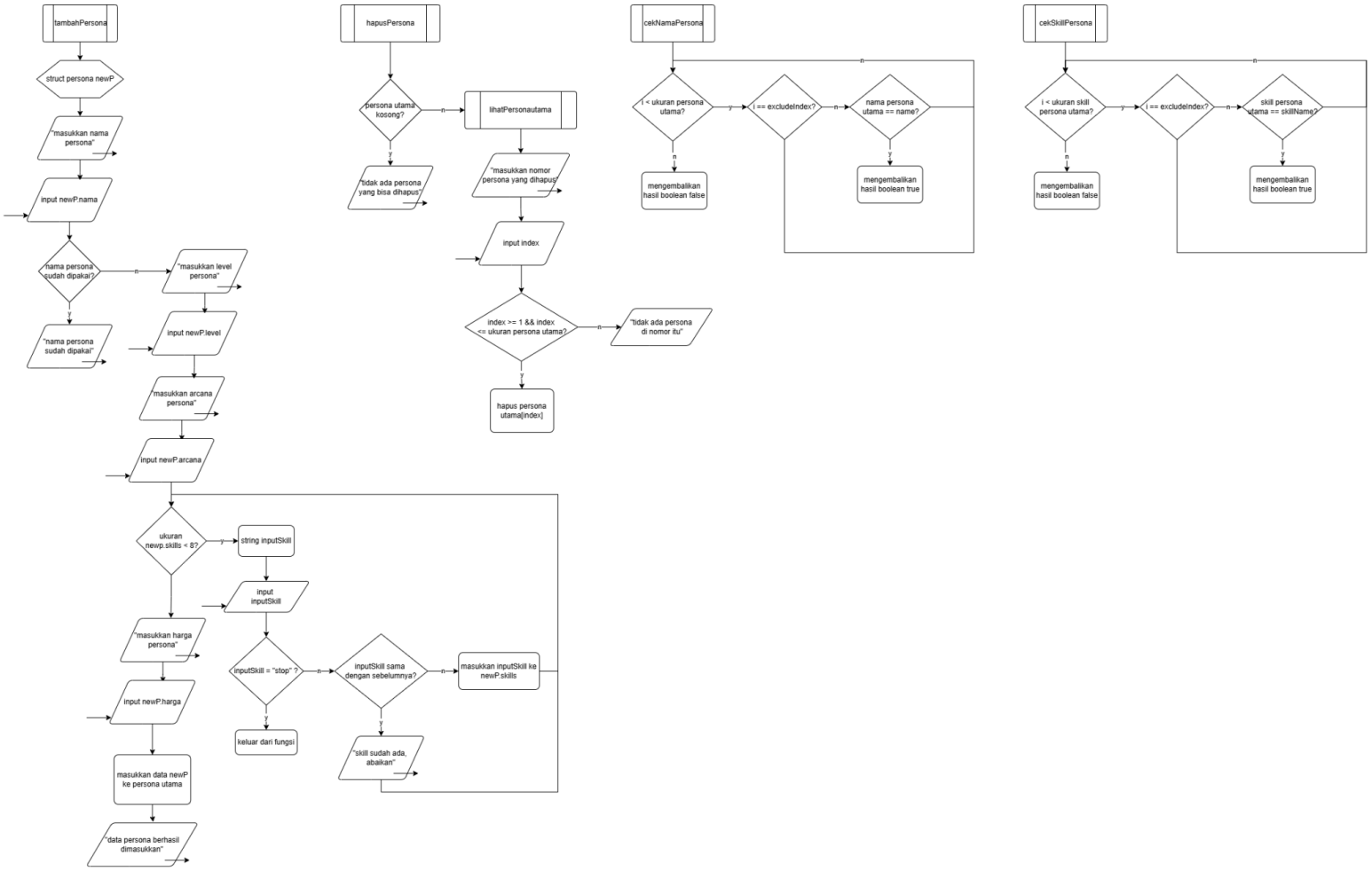
gambar 1.5 flowchart opsi user (fungsi beli dan lihat persona)



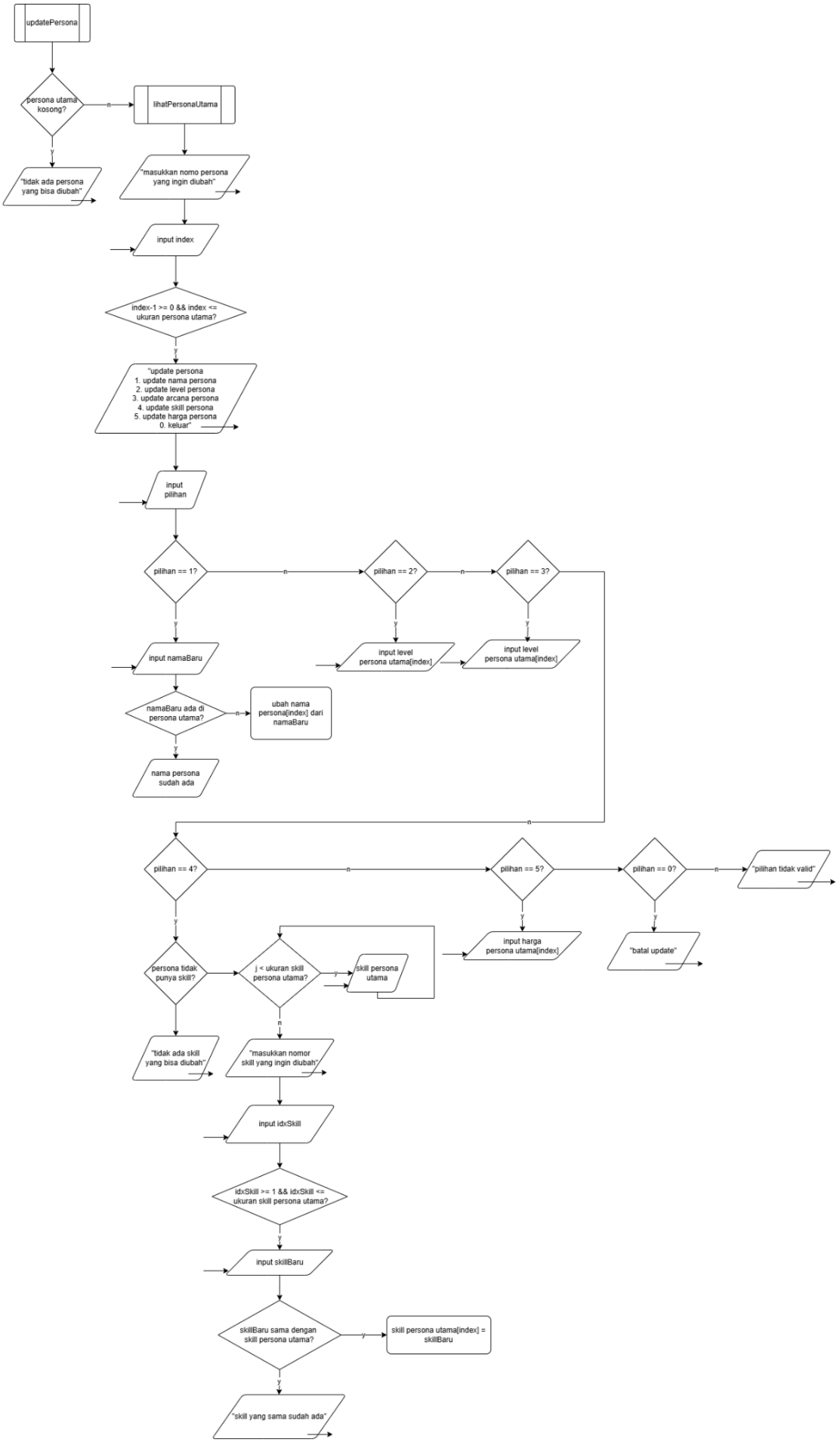
gambar 1.6 opsi user (fungsi gabung dan update persona)



gambar 1.7 opsi user dan admin (fungsi hapus persona user dan fungsi lihat persona utama)



gambar 1.8 opsi admin (fungsi tambah dan hapus persona utama)



gambar 1.9 opsi admin (update persona utama)

2. Deskripsi Singkat Program

pada posttest ini saya mencoba mengemas fitur fitur sebelumnya menjadi fungsi agar bisa dipakai berkali kali, saya juga menambahkan fitur untuk mengecek duplikasi agar tidak ada user/persona dengan nama yang sama

3. Source Code

A. fitur user

code ini terdapat semua fitur user, mulai dari lihat persona user hingga menghapus nya

Source Code:

```
else if(status == 2){
            percobaan = 0;
            string simpanNama = inputNama;
            int simpanIndeks = 0;
            if (totalDataUser == 0){
                dataPersonaUser[0].user = simpanNama;
                dataPersonaUser[0].jumlahPersona = 0;
                simpanIndeks = 0; //mastiin
                totalDataUser++;

            }else {
                int i = 0;
                while (i <= totalDataUser - 1)
                {
                    if(dataPersonaUser[i].user == simpanNama){
                        simpanIndeks = i;
                        i = 0;
                        break;
                    }
                    i++;
                }
            }
            int tempJumlah = dataPersonaUser[simpanIndeks].jumlahPersona;
            do {
                cout << "\n = menu velvet room user =" << endl;
```

```
                cout << "uang kamu : " << user[indeksUser].uang << " yen"<< endl;
                cout << "1. lihat data persona user" << endl;
                cout << "2. beli persona" << endl;
                cout << "3. fuse/gabung persona" << endl;
                cout << "4. ubah skill persona" << endl;
                cout << "5. hapus persona" << endl;
                cout << "0. keluar" << endl;
                cout << "Pilihan: ";
                cin >> pilihan;

                switch (pilihan)
                {
                    case 1:
                    if (dataPersonaUser[simpanIndeks].jumlahPersona == 0)
                    {
                        cout << "tidak ada persona yang dilihat" << endl;
                    }else {
                        for(int i = 0; i <=
dataPersonaUser[simpanIndeks].jumlahPersona - 1; i++){

                            cout << "\n ======" <<
dataPersonaUser[simpanIndeks].listPersona[i].nama << "======" << endl;
                            cout << "level    : "  <<
dataPersonaUser[simpanIndeks].listPersona[i].level << endl;
                            cout << "arcana   : "  <<
dataPersonaUser[simpanIndeks].listPersona[i].arcana << endl;

                            cout << "skill    : ";
                            for (int j = 0; j <
dataPersonaUser[simpanIndeks].listPersona[i].panjangSkill ; j++){
                                cout <<
dataPersonaUser[simpanIndeks].listPersona[i].skill[j] << ", ";
                            }
                            cout << endl;
                        }
                    }

                    break;
                    case 2:
                    for(int i = 0; i < totalPersona ; i++){
                        cout << "\n ======" << i + 1 << ". " << dataPersona[i].nama
<< "======" << endl;
                        cout << "level    : "  << dataPersona[i].level << endl;
                        cout << "arcana   : "  << dataPersona[i].arcana << endl;
                        cout << "skill    : ";
                        for (int j = 0; j < dataPersona[i].panjangSkill ; j++){
                            cout << dataPersona[i].skill[j] << ", ";
                        }
                        cout << endl;
                        cout << "harga   : "  << dataPersona[i].harga << endl;
                    }

                    cout << "pilih nomor persona yang ingin dibeli :"; cin >> index;
```

```
                    index -= 1;
                    if(index >= 0 && index <= totalPersona){
                        if(dataPersona[index].harga < user[indeksUser].uang){

dataPersonaUser[simpanIndeks].listPersona[tempJumlah].nama =
dataPersona[index].nama;

dataPersonaUser[simpanIndeks].listPersona[tempJumlah].level =
dataPersona[index].level;

dataPersonaUser[simpanIndeks].listPersona[tempJumlah].arcana =
dataPersona[index].arcana;
                            for(int i = 0; i <= dataPersona[index].panjangSkill - 1;
i++){

dataPersonaUser[simpanIndeks].listPersona[tempJumlah].skill[i] =
dataPersona[index].skill[i];
                            }

dataPersonaUser[simpanIndeks].listPersona[tempJumlah].panjangSkill =
dataPersona[index].panjangSkill;
                            dataPersonaUser[simpanIndeks].jumlahPersona++;
                            tempJumlah++;

                            user[indeksUser].uang -= dataPersona[index].harga;
                            cout << "persona berhasil dibeli" << endl;

                        }
                    }
                    break;
                    case 3:

                    if (dataPersonaUser[simpanIndeks].jumlahPersona < 2)
                    {
                        cout << "tidak ada persona yang bisa digabung" << endl;
                    }else {
                        for(int i = 0; i <=
dataPersonaUser[simpanIndeks].jumlahPersona - 1; i++){

                            cout << "\n ======" << i + 1 << ". "<<
dataPersonaUser[simpanIndeks].listPersona[i].nama << "======" << endl;
                            cout << "level    : "  <<
dataPersonaUser[simpanIndeks].listPersona[i].level << endl;
                            cout << "arcana   : "  <<
dataPersonaUser[simpanIndeks].listPersona[i].arcana << endl;

                            cout << "skill    : ";
                            for (int j = 0; j
<dataPersonaUser[simpanIndeks].listPersona[i].panjangSkill - 1; j++){
                                cout <<
dataPersonaUser[simpanIndeks].listPersona[i].skill[j] << ", ";
                            }
                    cout << endl;
```

```
                }
            }

            int persona1, persona2;

            cout << "masukkan persona pertama yang ingin digabung "; cin >>
persona1;
            cout << "masukkan persona kedua yang ingin digabung "; cin >> persona2;
            persona1 -= 1;
            persona2 -= 1;

            if((persona1 >= 0 && persona1 <
dataPersonaUser[simpanIndeks].jumlahPersona) && (persona2 >= 0 && persona2 <
dataPersonaUser[simpanIndeks].jumlahPersona) && persona1 != persona2 ){
                int target =
((dataPersonaUser[simpanIndeks].listPersona[persona1].level +
dataPersonaUser[simpanIndeks].listPersona[persona2].level) / 2 ) + 1;

                int terdekat = 0;
                double selisihTerkecil = abs(dataPersona[0].level);

                for (int i = 0; i < totalPersona ; i ++){
                    double selisihSekarang = abs(dataPersona[i].level - target);
                    if (selisihSekarang < selisihTerkecil) {
                        selisihTerkecil = selisihSekarang;
                        terdekat = i;
                    }
                }

                dataPersonaUser[simpanIndeks].listPersona[tempJumlah].nama =
dataPersona[terdekat].nama;
                dataPersonaUser[simpanIndeks].listPersona[tempJumlah].level =
dataPersona[terdekat].level;
                dataPersonaUser[simpanIndeks].listPersona[tempJumlah].arcana =
dataPersona[terdekat].arcana;
                    for(int i = 0; i <= dataPersona[terdekat].panjangSkill - 1;
i++){

dataPersonaUser[simpanIndeks].listPersona[tempJumlah].skill[i] =
dataPersona[terdekat].skill[i];
                    }

dataPersonaUser[simpanIndeks].listPersona[tempJumlah].panjangSkill =
dataPersona[terdekat].panjangSkill;
                    dataPersonaUser[simpanIndeks].jumlahPersona++;
                    tempJumlah++;

                    int maxIdx = max(persona1, persona2);
                    int minIdx = min(persona1, persona2);

                    for(int i = maxIdx; i < tempJumlah - 1 ; i++){
                        dataPersonaUser[simpanIndeks].listPersona[i] =
dataPersonaUser[simpanIndeks].listPersona[i + 1];
```

```
                        dataPersonaUser[simpanIndeks].jumlahPersona--;
                        tempJumlah--;
                    }

                    for(int i = minIdx; i < tempJumlah - 1 ; i++){
                        dataPersonaUser[simpanIndeks].listPersona[i] =
dataPersonaUser[simpanIndeks].listPersona[i + 1];
                        dataPersonaUser[simpanIndeks].jumlahPersona--;
                        tempJumlah--;
                    }

                }else {
                    cout << "nomor yang dipilih tidak ada" << endl;
                }

                break;
                case 4:
                if(dataPersonaUser[simpanIndeks].jumlahPersona == 0){
                    cout << "kamu belum punya persona apa apa" << endl;
                    break;
                }
                for(int i = 0; i <= tempJumlah - 1; i++){
                    cout << i + 1 << ". " <<
dataPersonaUser[simpanIndeks].listPersona[i].nama << endl;
                }
                cout << "pilih nomor persona : "; cin >> index;
                index -= 1;
                if (index >= 0 && index <= tempJumlah - 1) {
                    int tempSkill =
dataPersonaUser[simpanIndeks].listPersona[index].panjangSkill;
                        for(int j = 0; j < tempSkill ; j ++){
                            cout << j + 1 << ". " <<
dataPersonaUser[simpanIndeks].listPersona[index].skill[j] << endl;
                        }

                    cout << "pilih skill yang ingin diubah "; cin >> indexSkill;

                    if(indexSkill - 1 >= 0 && indexSkill <= tempSkill){
                        int indexSkillItem;
                        for(int i = 0; i < 5; i++ ){
                            cout << i + 1 << ". "<< skillItem[i] << " " << endl;;
                        }
                        cout << "pilih nomor skill baru : "; cin >> indexSkillItem;

                        if(indexSkillItem - 1 >= 0 && indexSkillItem <= 5){
                                for(int j = 0; j <= tempSkill - 1; j ++){

if(dataPersonaUser[simpanIndeks].listPersona[index].skill[j] ==
skillItem[indexSkillItem - 1]){
                                        jumlahDataSama = true;
                                    }
                                }
                                if(jumlahDataSama){
```

```
                                    cout << "sudah punya skill yang sama, batalkan
pembaruan" << endl;
                                }else{

dataPersonaUser[simpanIndeks].listPersona[index].skill[indexSkill -1] =
skillItem[indexSkillItem - 1];
                                    cout << "pembaruan berhasil" << endl;
                                }
                            }else {
                        cout << "nomor skill item tidak ada" << endl;
                    }
                        }else {
                        cout << "nomor skill Persona tidak ada" << endl;
                    }
                    }else {
                        cout << "nomor Persona tidak ada" << endl;
                    }
                    break;
                    case 5:
                    if(dataPersonaUser[simpanIndeks].jumlahPersona == 0){
                        cout << "kamu belum punya persona apa apa" << endl;
                        break;
                    }else{
                        for(int i = 0; i <= dataPersonaUser[simpanIndeks].jumlahPersona
- 1; i++){

                            cout << "\n ======" << i + 1 << ". "<<
dataPersonaUser[simpanIndeks].listPersona[i].nama << "======" << endl;
                            cout << "level    : "  <<
dataPersonaUser[simpanIndeks].listPersona[i].level << endl;
                            cout << "arcana   : "  <<
dataPersonaUser[simpanIndeks].listPersona[i].arcana << endl;

                            cout << "skill    : ";
                            for (int j = 0; j
<dataPersonaUser[simpanIndeks].listPersona[i].panjangSkill - 1; j++){
                                cout <<
dataPersonaUser[simpanIndeks].listPersona[i].skill[j] << ", ";
                            }
                        }
                    }
                    cout << "pilih persona yang ingin dihapus" ; cin >> index;
                    if (index - 1 >= 0 && index - 1 <= tempJumlah - 1 ){
                        for(int i = index - 1; i < tempJumlah - 1; i++){
                            dataPersonaUser[simpanIndeks].listPersona[i] =
dataPersonaUser[simpanIndeks].listPersona[i + 1];
                        }
                    }else{
                        cout << "nomor persona tidak ada" << endl;
                    }
                    break;
                    case 0:
                    cout << "log out" << endl;
```

```
                status = 0;
                break;

                default:
                cout << "Pilihan tidak valid." << endl;
                break;
            }
        }while(pilihan != 0);
```

B. Fitur admin

pada kode ini terdapat semua fitur admin, mulai dari lihat data utama persona, hingga hapus persona dari data utama

```
if( status == 1){

            percobaan = 0;
            do {
              cout << "\n = menu velvet room admin =" << endl;
              cout << "1. tampilkan data persona" << endl;
              cout << "2. Tambah persona " << endl;
              cout << "3. update persona" << endl;
              cout << "4. Hapus persona" << endl;
              cout << "0. Keluar" << endl;
              cout << "Pilihan: ";
              cin >> pilihan;
              switch (pilihan) {
                case 1:
                if (totalPersona == 0){
                   cout << "tidak ada persona yang bisa dibaca" << endl;
                }else{
                for(int i = 0; i < totalPersona ; i++){
                    cout << "\n ======" << dataPersona[i].nama << "======" << endl;
                    cout << "level    : "  << dataPersona[i].level << endl;
                    cout << "arcana   : "  << dataPersona[i].arcana << endl;
                    cout << "skill    : ";
                    for (int j = 0; j < dataPersona[i].panjangSkill ; j++){
                        cout << dataPersona[i].skill[j] << ", ";
                    }
                    cout << endl;
                    cout << "harga   : "  << dataPersona[i].harga << endl;
                }
            }

                break;

                case 2:

                if (totalPersona < 20){
                    cout << "masukkan nama persona : ";
```

```
                    cin.ignore();
                    getline(cin, dataPersona[totalPersona].nama);

                    cout << "masukkan level persona : ";
                    cin.ignore();
                    cin >> dataPersona[totalPersona].level;

                    cout << "masukkan arcana persona : ";
                    cin.ignore();
                    cin >> dataPersona[totalPersona].arcana;

                    cout << "masukkan skill persona : " << endl;
                    cout << "ketik stop untuk berhenti : " << endl;
                    cin.ignore(1000, '\n');
                    int i = 0;
                    while(i <= 7){
                        string inputSkill;
                        cout << "skill " << i + 1 << ": "; getline(cin,inputSkill);

                        if (inputSkill == "stop"){
                        break;
                    }

                        dataPersona[totalPersona].skill[i] = inputSkill;
                        dataPersona[totalPersona].panjangSkill++;
                        i++;
                    }

                    cout << "masukkan harga persona : ";
                    cin.ignore();
                    cin >> dataPersona[totalPersona].harga;
                    totalPersona++;
                    cout << "data berhasil dimasukkan" << endl;
                }else{
                    cout << "kapasitas data penuh" << endl;
                }
                break;

                //update
                case 3:

                if (totalPersona == 0){
                   cout << "tidak ada persona yang bisa diubah" << endl;
                }else{
                    for(int i = 0; i < totalPersona ; i++){
                        cout << "persona ke " << i + 1 << " : " <<
dataPersona[i].nama << endl;
                    }
                    cout << "masukkan nomor persona yang ingin diubah" << endl ;
                    cin >> index;

                    if(index >= 1 && index <= totalPersona){
                        do {
```

```
                            cout << "1. ubah nama persona" << endl;
                            cout << "2. ubah level persona " << endl;
                            cout << "3. ubah arcana persona" << endl;
                            cout << "4. ubah skill persona" << endl;
                            cout << "5. ubah  harga persona" << endl;
                            cout << "0. batal update" << endl;
                            cout << "Pilihan: ";
                            cin >> pilihan;
                            switch (pilihan)
                            {
                            case 1:
                            cout << "masukkan nama baru : ";
                            cin.ignore();
                            cin >> dataPersona[index - 1].nama;
                            cout << "berhasil diubah" << endl;
                            break;
                            case 2:
                            cout << "masukkan level baru : ";
                            cin.ignore();
                            cin >> dataPersona[index - 1].level;
                            cout << "berhasil diubah" << endl;
                            break;
                            case 3:
                            cout << "masukkan arcana baru : ";
                            cin.ignore();
                            cin >> dataPersona[index - 1].arcana;
                            cout << "berhasil diubah" << endl;
                            break;
                            case 4:
                            for (int j = 0; j < dataPersona[index - 1].panjangSkill
; j++){
                                cout << "skill ke " << j + 1 << " : " <<
dataPersona[index - 1].skill[j] << endl;
                            }
                            cin.ignore();
                            cout << "masukkan nomor skill yang ingin diubah :"; cin
>> indexSkill;
                            indexSkill -= 1;
                            if(indexSkill >= 0 && indexSkill <= dataPersona[index -
1].panjangSkill - 1){
                                cin.ignore();
                                getline(cin,dataPersona[index -
1].skill[indexSkill]);
                                cout << "skill berhasil diubah" << endl;
                            }
                            break;
                            case 5:
                            cout << "masukkan harga baru : ";
                            cin.ignore();
                            cin >> dataPersona[index - 1].harga;
                            cout << "berhasil diubah" << endl;
                            break;
```

```
                            case 0:
                            cout << "batal update" << endl;
                            break;

                            default:
                            cout << "nomor tidak valid";
                            break;
                            }
                        } while(pilihan != 0);
                    }else {
                        cout << "tidak ada persona di nomor itu" << endl;
                    }
                }

                break;
                case 4:
                if (totalPersona == 0){
                   cout << "tidak ada persona yang bisa dihapus" << endl;
                }else{
                    for(int i = 0; i < totalPersona ; i++){
                        cout << "persona ke " << i + 1<< " : " <<
dataPersona[i].nama << endl;
                    }
                    cout << "masukkan nomor persona yang ingin dihapus" << endl ;
                    cin >> index;

                    if(index >= 1 && index <= totalPersona){
                        for(int i = index - 1 ; i < totalPersona - 1; i++){
                           dataPersona[i] = dataPersona[i + 1];
                        }
                        totalPersona--;
                        cout << "persona berhasil dihapus" << endl;
                    }else {
                        cout << "tidak ada persona di nomor itu" << endl;
                    }
                }
                break;
                case 0:
                cout << "log out...." << endl;
                status = 0;
                break;
                default:
                cout << "Pilihan tidak valid." << endl;
                }
            } while (pilihan != 0);
```

4. Hasil Output

```
pilihan : 2

====== 1orpheus ======
Level     : 10
Arcana    : fool
Skill     : agi, dia, bash, tarunda,
Harga     : 500

====== 2jack o lantern ======
Level     : 15
Arcana    : magician
Skill     : bufu, pulinpa, mabufu, shock punch, freeze boost, magic boost,
Harga     : 750

====== 3archangel ======
Level     : 20
Arcana    : justice
Skill     : kouha, single shot, baisudi, sukukaja, holy arrow, apt pupil,
Harga     : 1000
Pilih nomor persona yang ingin dibeli (0 untuk batal) : 3
persona berhasil dibeli! Sisa uang: 9000
```

```
====== 1archangel ======
Level     : 20
Arcana    : justice
Skill     : kouha, single shot, baisudi, sukukaja, holy arrow, apt pupil,
```

Gambar 4.1 opsi user(beli persona)

```
pilihan : 3

====== 1jack o lantern ======
Level     : 15
Arcana    : magician
Skill     : bufu, pulinpa, mabufu, shock punch, freeze boost, magic boost,

====== 2archangel ======
Level     : 20
Arcana    : justice
Skill     : kouha, single shot, baisudi, sukukaja, holy arrow, apt pupil,
masukkan nomor persona pertama : 1
masukkan nomor persona kedua : 2
Fusion berhasil! Mendapatkan: trilionos ngutangopoulos
```

```
====== 1trilionos ngutangopoulos ======
Level     : 17
Arcana    : aeon
Skill     : 11 ribu triliun, jalan tol,
```

Gambar 4.2 opsi user (fuse/gabung persona)

```
pilihan : 4
1. trilionos ngutangopoulos
Pilih nomor persona : 1
1. 11 ribu triliun
2. jalan tol
Pilih skill yang ingin diubah : 2
Daftar Skill Item:
1. masukunda
2. dekaja
3. debilitate
4. charge
5. concentrate
Pilih nomor skill baru : 5
Pembaruan berhasil
```

```
====== 1trilionos ngutangopoulos ======
Level     : 17
Arcana    : aeon
Skill     : 11 ribu triliun, concentrate,
```

gambar 4.3 opsi user (ubah skill persona)

```
pilihan : 5

====== 1trilionos ngutangopoulos ======
Level     : 17
Arcana    : aeon
Skill     : 11 ribu triliun, concentrate,

====== 2archangel ======
Level     : 20
Arcana    : justice
Skill     : kouha, single shot, baisudi, sukukaja, holy arrow, apt pupil,
Pilih persona yang ingin dihapus : 1
persona berhasil dihapus
```

```
pilihan : 1

====== 1archangel ======
Level     : 20
Arcana    : justice
Skill     : kouha, single shot, baisudi, sukukaja, holy arrow, apt pupil,
```

gambar 4.4 opsi user (hapus persona user)

```
pilihan: 2
masukkan nama persona : trilionos ngutangopoulos
masukkan level persona : 17
masukkan arcana persona : aeon
masukkan skill persona (ketik 'stop' untuk berhenti):
Skill 1: 11 ribu triliun
Skill 2: jalan tol
Skill 3: stop
masukkan harga persona : 500
Data persona berhasil dimasukkan
```

```
====== 4trilionos ngutangopoulos ======
Level     : 17
Arcana    : aeon
Skill     : 11 ribu triliun, jalan tol,
Harga     : 500
```

gambar 4.5 opsi admin (tambah persona)

```
--- Update persona: trilionos ngutangopoulos ---
1. Ubah nama persona
2. Ubah level persona
3. Ubah arcana persona
4. Ubah skill persona
5. Ubah harga persona
0. Batal update
pilihan: 4
Skill ke 1 : 11 ribu triliun
Skill ke 2 : jalan tol
masukkan nomor skill yang ingin diubah : 2
masukkan skill baru : ijazah palsu
Skill berhasil diubah
```

```
====== 4trilionos ngutangopoulos ======
Level     : 17
Arcana    : aeon
Skill     : 11 ribu triliun, ijazah palsu,
Harga     : 500
```

gambar 4.6 opsi admin (edit persona)

```
====== 3archangel ======
Level     : 20
Arcana    : justice
Skill     : kouha, single shot, baisudi, sukukaja, holy arrow, apt pupil,
Harga     : 1000

====== 4trilionos ngutangopoulos ======
Level     : 17
Arcana    : aeon
Skill     : 11 ribu triliun, ijazah palsu,
Harga     : 500
masukkan nomor persona yang ingin dihapus : 3
persona berhasil dihapus
```

```
====== 1orpheus ======
Level     : 10
Arcana    : fool
Skill     : agi, dia, bash, tarunda,
Harga     : 500

====== 2jack o lantern ======
Level     : 15
Arcana    : magician
Skill     : bufu, pulinpa, mabufu, shock punch, freeze boost, magic boost,
Harga     : 750

====== 3trilionos ngutangopoulos ======
Level     : 17
Arcana    : aeon
Skill     : 11 ribu triliun, ijazah palsu,
Harga     : 500
```

gambar 4.7 opsi admin(hapus persona)

```
=== SELAMAT DATANG DI VELVET ROOM ===
1. login
2. register
0. keluar
pilihan : 2

=== Registrasi user baru ===
masukkan nama user : sawitLover
masukkan password : heiAntekAntek4sin99
registrasi berhasil
```

```
--- Menu Login ---
masukkan nama (atau 0 untuk keluar): sawitLover
masukkan Password : heiAntekAntek4sin99
selamat datang sawitLover

= Menu Velvet Room User =
User: sawitLover | Uang: 10000 yen
1. Lihat data persona user
2. Beli persona
3. Fuse/Gabung persona
4. Ubah skill persona
5. Hapus persona
0. Keluar
```

gambar 4.8 register user

5. Langkah-langkah GIT

5.1 GIT Init

```
ozora@ozra MINGW64 /g/praktikum/APL (main)
$ git init
Reinitialized existing Git repository in G:/praktikum/APL/.git/
```

gambar 5.1 inisiasi git

5.2 GIT Add

```
ozora@ozra MINGW64 /g/praktikum/APL (main)
$ git add /g/praktikum/APL/kelas/post-test/post-test-apl-3/2509106038-FEBRIANNO-
OZORA-ALINANTO-PT-3.exe /g/praktikum/APL/kelas/post-test/post-test-apl-3/2509106
038-FEBRIANNO-OZORA-ALINANTO-PT-3.cpp
```

gambar 5.2 menambahkan file baru ke git

5.3 GIT Commit

```
ozora@ozra MINGW64 /g/praktikum/APL (main)
$ git commit -m "bunch of mess part 2"
[main c7e4cae] bunch of mess part 2
 2 files changed, 581 insertions(+)
 create mode 100644 kelas/post-test/post-test-apl-3/2509106038-FEBRIANNO-OZORA-A
LINANTO-PT-3.cpp
 create mode 100644 kelas/post-test/post-test-apl-3/2509106038-FEBRIANNO-OZORA-A
LINANTO-PT-3.exe
```

gambar 5.3 commit git local

5.4 GIT Remote

```
ozora@ozra MINGW64 /g/praktikum/APL (main)
$ git remote add origin https://github.com/ojola444/praktikum-apl/git
error: remote origin already exists.
```

gambar 5.4 git remote

5.5 GIT Push

```
ozora@ozra MINGW64 /g/praktikum/APL (main)
$ git push -u origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 16 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 154.91 KiB | 8.15 MiB/s, done.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/ojola444/praktikum-apl.git
   810b6ff..c7e4cae  main -> main
branch 'main' set up to track 'origin/main'.
```

gambar 5.5 push commit local ke repository main

5.7 hasil push github